

Mini Project 1: Wind Energy Converter

General: your input data is given in the assignment sheet “AS” and is based on your personal number.

A. MP1 examination is in the form of a presentation, which should not be more than 3 slides.

B. Presentation:

The results should be presented to the group on a zoom-seminar, book 1 h in group set “MP1 Presentation” in Studium/people. Present dimensioned WEC and aspects of annual energy production, mechanics of tower and plausible driveline (results from point 1-5 in instruction below),

Comment on your WEC and the energy production – anything bad/good/strange/suboptimal?? Presentation should be maximum 3 slides and maximum 3 minutes long and allow for 2 min questions and comments (6 students per 45 min session).

C. Focus presentation on:

Sticking to the time limit -3 min, a short intro with given background data, present results (as of point 2 above), present some reflections and comments (on the turbine, the results, the task and what was difficult...).

1. Wind Resource

Your wind resource is given by a Weibull distribution with the shape factor “ k ” and average wind speed “ $\bar{U}$ ” [m/s] @ 10m elevation given (see Assignment Sheet “AS”).

Calculate average wind speed at hub height (use H_h from WEC below) U_h using a surface roughness length in [mm] for terrain with z_0 from AS. Comment on what sort of terrain this could represent?

Use k and U_h to calculate the number of expected hours for each wind speed in bins using **integers** for wind speed (i.e. for 0, 1, 2, 3... m/s). Assume that your downtime $X\%$ [%] (AS) is evenly distributed over wind speed bins.

Plot the annual distribution of hours at each wind speed AND the distribution of wind energy (per square meter [kWh/m²]) for the wind speeds, in the same plot (with different scales so the peaks in the plot is about the same height)

Rated wind speed: select wind speed (integer) that gives cumulative energy/sqm of a 1/3 of total resource i.e. make sure that at least 1/3 of the **wind energy resource** is below the selected rated wind speed (using your plot of distribution of wind energy as dependent on wind speed).

*Example: There is 5 kWh/sqm/slot per year available for 3, 4 and 5 m/s; 10 kWh/sqm/slot for 6, 7, 8 and 9 AND 3 kWh/sqm/slot for 10 and 11 m/s respective wind slots. Thus, the total resource is $3*5 + 4*10 + 2*3 = 61$ kWh/sqm/year. A 1/3 of this is 20.66 kWh/sqm. Thus, at 7 m/s more than a 1/3 of the resource is below (actually 25 kWh/sqm). Thus, rated wind should be chosen to 7 m/s.*

Select cut-out wind speed (integer) that corresponds cumulative energy/sqm of at least 80% of total resource below it; i.e. in our *example* that cut-out would be chosen to 10 m/s (meaning production for 10 m/s = 0 kWh).

2. Wind Energy Converter ‘WEC’

AS gives your: wind turbine diameter D_T , tower height H_h (hub height), the C_p of your turbine AND the **combined efficiency** of your driveline, generator and grid connection is η_{tot} . (i.e. is $\eta_{tot} = \eta_d * \eta_g * \eta_c$)

Calculate **Rated Power** (i.e. electric rated power, use C_p and η_{tot}) and present it.

Select **cut-in** wind speed (one decimal) for a power level, which is one 1% of rated power.

Present **these results** in a plot of the power curve AND print number of selection above.

3. Energy production for one year

Calculate the total generated electric energy of your turbine for one year (in MWh), average power and corresponding number of full-load-hours (using bins of above).

Check1: calculate full-load-hours (which should be well above total number of hours at rated power).

(OBS the term “full-load-hours” does not refer to the actual number of hours the turbine generates full power. Instead, it relates to the full energy production over the year –make sure you understand this term before answering i.e. calculate the number of equivalent full load hours).

Check2: Calculate the average power production (over the year) for a wind speed higher than rated but lower than cut-out, by dividing the energy you calculated (for this wind speed) with the number of hours at this speed (cannot be higher than rated power).

Plot the produced energy as function of bins (wind speed in integers) – compare to the annual distribution of available wind energy (per bin).

4. Mechanics of tower

Dimension a cylindrical tower of steel with $\sigma_{\text{allowed}} = 160 \text{ MPa}$ and a diameter of $H_r/20 \text{ [m]}$. Tower top mass is estimated with a nacelle weight in respect to rated power of 40 kg/kW .

Consider 3 load cases, calculate required wall thickness for each (and present these values):

a) Gravity load on base of tower from nacelle and tower weight.

b) Wind load on turbine at rated power. Add a factor of 2 for wall thickness to compensate for buckling (which is too complicated to calculate for this assignment).

c) Extreme wind load on tower from turbine and tower for an IEC class II turbine (60 m/s). Add a factor of 2 for wall thickness to compensate for buckling.

d) Select wall thickness based on the most severe case (a-c above).

TIP: Make sure your values **are realistic**: don't answer with a wall thickness of several meters or something equivalent to the thickness of human hair...

5. Efficiency of driveline

Comment on the efficiency of your WEC: how it could be distributed such that total efficiency (η_{tot} , given in AS) is a product of each (i.e. $\eta_{\text{tot}} = \eta_{\text{driveline}} * \eta_{\text{generator}} * \eta_{\text{gridconnection}}$). You need to argue for the driveline you have, what generator you have selected et cetera –such that the numbers are somewhat realistic.

For example: a PM synchronous generator with an efficiency of 0.72 is not realistic (you should know why)...

6. Presentation:

The results should be presented to the group on a zoom seminar (book a group in Studium “MP1 Presentation”). Present the dimensioned WEC and aspects of annual energy production, mechanics of tower and plausible driveline (point 1-5 above).

Comment on you WEC and the energy production – anything bad/good/strange/suboptimal???

Presentation should be maximum 3 slides, maximum 3 minutes long, and 2 min for questions leaving time for changing presenter (6 students per 45 min).

Before presentation slides must be submitted in PDF-format in Assignment MP1 (showing all results asked for above). Make sure to update it with respect to presentation feedback. If additional revision are required a new presentation must be resubmitted in Assignment MP1.

😊 Good Luck!